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MASTER DRAINAGE PLAN

Cotton Creek Ranch, Rev 2

Prepared by Halvard Water Resources

2026-07-15

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REVISION SUMMARY

Updated to Land Plan Rev C impervious cover; Pond 2 moved east of the wetlands; total storage raised to 688 ac-ft.

2.0 EXISTING WATERSHEDS

The tract drains to Cotton Creek through Tributaries 1 to 4.

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3.0 HYDROLOGY

Runoff was computed for the 2-, 10-, 100- and 500-year events.

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EXHIBIT

Drainage area map (graphic not reproduced in text).

APPENDIX

Model input and output (not reproduced in text).

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About 210 acres lie within the effective 1 percent annual chance floodplain, including 64 acres of floodway.

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Impervious cover of 56 percent was assumed from Land Plan Revision C.

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The three regional detention ponds provide a combined storage volume of 688 acre-feet.

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Pond 2 is relocated east of the delineated wetlands along the Phase 2 boundary.

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Peak discharge to Cotton Creek shall not exceed the existing 100-year peak of 1,240 cfs.

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Minimum slab elevations shall be set 2 feet above the ultimate 1 percent water surface elevation.

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Approximately 2.4 miles of Tributary 3 will be improved with 4 horizontal to 1 vertical side slopes.

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